ASSESSMENT TOOL 1

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

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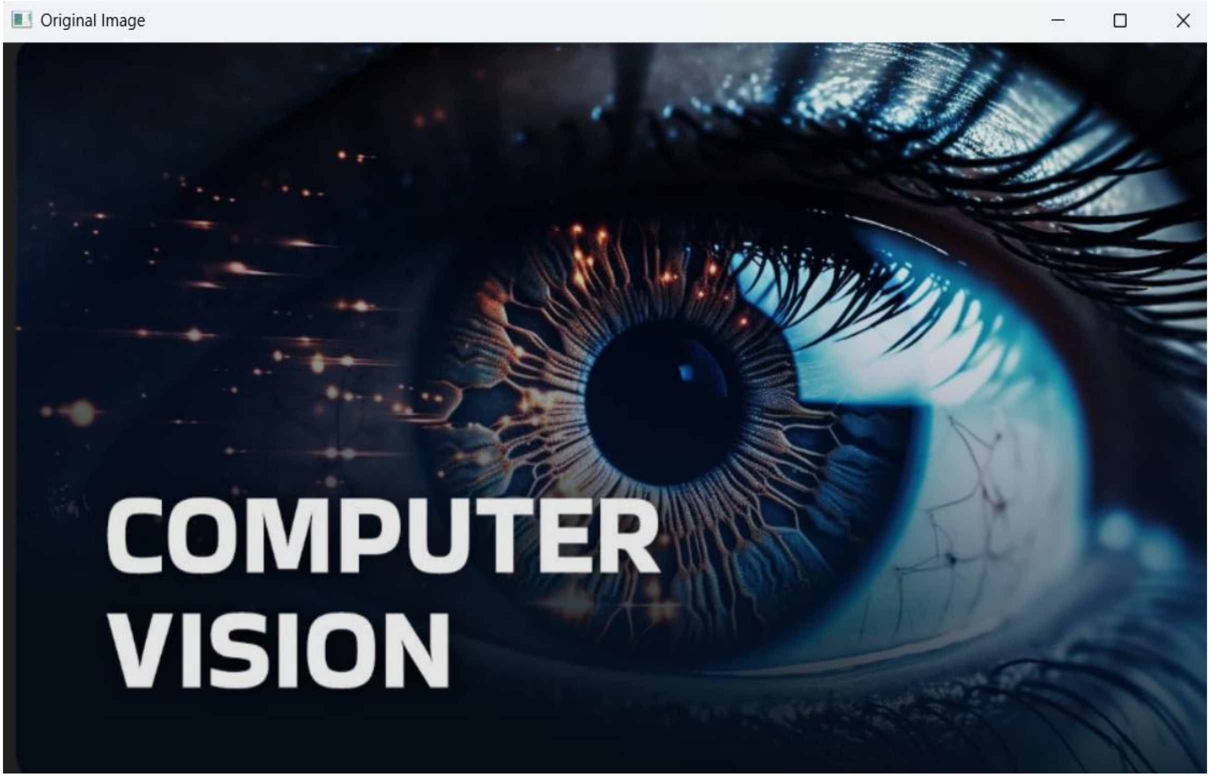
1.A camera captures an image that appears blurred due to improper sampling.

Write a Python program to demonstrate image sampling.

Sampling.py - C:/Users/Yavan/AppData/Local/Programs/Python/Python314/Sampling.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2

# Read image
img = cv2.imread(r"C:\Users\Yavan\OneDrive\Pictures\Screenshots\Screenshot 2026-07-27 203351.png")

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
    exit()

# Downsample - reduce sampling rate (take fewer pixels)
small = cv2.resize(img, (img.shape[1]//8, img.shape[0]//8), interpolation=cv2.INTER_NEAREST)

# Upsample back - shows blur/blockiness from improper sampling
sampled = cv2.resize(small, (img.shape[1], img.shape[0]), interpolation=cv2.INTER_NEAREST)

# Display
cv2.imshow("Original Image", img)
cv2.imshow("Improper Sampling Result", sampled)

cv2.waitKey(0)
cv2.destroyAllWindows()
```


Original Image
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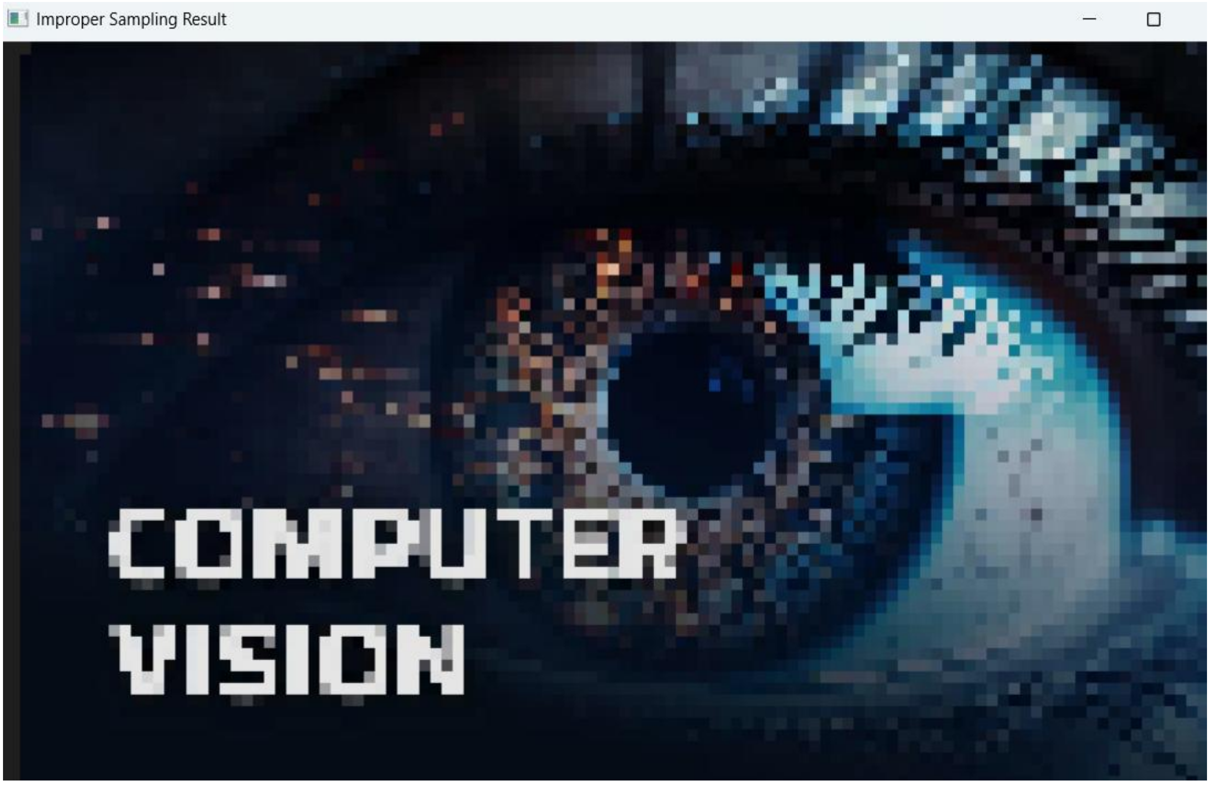
Improper Sampling Result
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2.A real-world scene is captured under low-light conditions.

Write a Python program to enhance the low-light image.

Lowlight.py - C:/Users/Yavan/AppData/Local/Programs/Python/Python314/Lowlight.py (3.14.0)

File Edit Format Run Options Window Help

```python
import cv2

# Read image
img = cv2.imread(r"C:\Users\Yavan\OneDrive\Pictures\Screenshots\Screenshot 2026-07-27 210606.png")

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
    exit()

# Convert to LAB color space
lab = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)
l, a, b = cv2.split(lab)

# Apply CLAHE to the L-channel (brightness) only
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
l_enhanced = clahe.apply(l)

# Merge back and convert to BGR
enhanced_lab = cv2.merge((l_enhanced, a, b))
enhanced = cv2.cvtColor(enhanced_lab, cv2.COLOR_LAB2BGR)

# Display
cv2.imshow("Original Low-Light Image", img)
cv2.imshow("Enhanced Image", enhanced)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

3.An automated surveillance system monitors moving objects.

Write a Python program for motion detection.

File Edit Format Run Options Window Help

```
import cv2

# Start webcam
cap = cv2.VideoCapture(0)

# Read first two frames
ret, frame1 = cap.read()
ret, frame2 = cap.read()

while cap.isOpened():
    # Find difference between frames
    diff = cv2.absdiff(frame1, frame2)
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
    blur = cv2.GaussianBlur(gray, (5, 5), 0)
    _, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)
    dilated = cv2.dilate(thresh, None, iterations=3)

    # Find contours of moving objects
    contours, _ = cv2.findContours(dilated, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)

    for c in contours:
        if cv2.contourArea(c) < 900:
            continue
        x, y, w, h = cv2.boundingRect(c)
        cv2.rectangle(frame1, (x, y), (x+w, y+h), (0, 255, 0), 2)
        cv2.putText(frame1, "Motion Detected", (10, 30), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)

    # Display
    cv2.imshow("Motion Detection", frame1)

    # Move to next frame
    frame1 = frame2
    ret, frame2 = cap.read()

    if cv2.waitKey(10) == 27:  # Press ESC to quit
        break

cap.release()
cv2.destroyAllWindows()
```


Motion Detection
Motion Detected